DREW T ERNST

DATA ANALYST | SOFTWARE ENGINEER | PROGRAM DESIGNER

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Strategic Data Analyst and Software Engineer and Developer with a foundation in Public Relations and Marketing, specializing in the intersection of Data Integrity and high-fidelity application architecture. Expert in clinical narrative validation, behavioral financial modeling, and modular system design. Proven track record of delivering "Zero-Footprint" logic engines and automated deployment pipelines that satisfy strict compliance and audit standards in high-stakes environments.

TECHNICAL COMPETENCIES

Data Analysis & Integrity:
Data Integrity, Categorical to Ordinal Encoding, Grouped Imputation, Correlation vs. Causation Analysis, SQL.

Backend & APIs:
Python (FastAPI), Node.js, Express.js, MongoDB, RESTful APIs, JWT Authentication.

Frontend Engineering:
React, Vite, TypeScript, Vanilla JavaScript (ES6+), Tailwind CSS, GSAP Animation.

Architecture & DevOps:
Docker, Nginx, GitLab CI/CD, AWS (S3, Lambda), Version Control (Git).

DATA ENGINEERING INITIATIVES AND CASE STUDIES

MCSDD | PCSP Assistant Pro (Clinical Case Study)

Missouri State Agency Compliance

Case Study: https://dte-solutions.icu/case-study-pcsp/

→ **Architected a Zero-Footprint Logic Engine** ensuring 100% **Data Integrity** for HIPAA-compliant clinical narratives, eliminating plan rejections for state agencies.

→ Developed client-side validation logic that reduced documentation revision cycles by 40%, satisfying strict Missouri DMH audit standards through precise clinical narrative validation.

→ Applied **Correlation vs. Causation** analysis to service reporting, ensuring accurate clinical outcomes were reflected in government-mandated documentation.

Fluff | Swing Coach, Stats Tracker and Golf Assistant

Architecture & Performance

→ Engineered a high-performance tracking platform using **Vite and React**, utilizing a modularized, flattened architecture for rapid scaling and maintainability.

→ Architected and deployed a containerized environment using **Docker and Nginx**, integrated with a **GitLab CI/CD** pipeline for automated, practice-ready delivery.

→ Implemented **Categorical to Ordinal** encoding to transform player performance metrics into ranked benchmarking data for statistical analysis.

Pulse | Ai Driven Behavioral Financial Tracker & Assistant

Full-Stack Data Science

→ Developed a behavioral finance dashboard featuring a **9-step behavioral onboarding flow**, optimizing data collection and user engagement. Zero legacy bloat; built to scale on modern infrastructure.

- → Cloud Architecture: AWS S3 integration creates "sticky" workflows.
- → Trends Engine, Predictive spending habits, Actionable insights
- → Real-time net worth / budget, Dark-mode, ultra-fast UI, Encrypted, searchable vault.
- → Built a robust **Python/FastAPI** backend with MongoDB, utilizing **Grouped Imputation** techniques to maintain accuracy in multi-period wealth forecasting models.

PROFESSIONAL EXPERIENCE

Data Analyst & Software Engineer | Developer

Self-Employed | 2025 – Present

- → Engineer custom, data-driven web applications with a focus on component-based architecture and rigorous data validation logic.
- → Consult on process optimization and data integrity for state agencies and local service businesses, bridging technical execution with business strategy.

EDUCATION & LEADERSHIP

Frontend Simplified Bootcamp & Mentorship — **Mentored by David Bragg | 2025**

Bachelor of Arts in Public Relations — **Quincy University | GPA: 3.8 | 2011-2012**

Bachelor of Science in Marketing — **Culver-Stockton College | GPA: 3.8 | 2002–2005**

S.I.F.E (Students in Free Enterprise)